

A photograph of a modern university building at night, with its interior lights glowing through large windows and reflecting in a body of water in the foreground. The sky is a deep purple.

AGILE SOFTWARE ENGINEERING:

AGILE MANIFESTO & SCRUM

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Last lecture's objectives

Overview of the content and activities in the agile software engineering (ASE) course.

Understand the software engineering problem and that it has fundamentally different answers.

Knowledge of different paradigms for organizing software engineering (process models).

Lecture objectives

A historical overview of the agile manifesto and scrum

Knowledge about scrum roles, events, and artifacts

Skills in using scrum in a software project

Manifesto for Agile Software Development

We are uncovering better ways of developing software by doing it and helping others do it.
Through this work we have come to value:

Individuals and interactions over processes and tools

Working software over comprehensive documentation

Customer collaboration over contract negotiation

Responding to change over following a plan

That is, while there is value in the items on the right, we value the items on the left more.

What may be the problems these 4 values should solve?

Principles behind the Agile Manifesto

- **Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.**
- **Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.**
- **Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the shorter timescale.**
- **Business people and developers must work together daily throughout the project.**
- **Build projects around motivated individuals. Give them the environment and support they need, and trust them to get the job done.**
- **The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.**
- **Working software is the primary measure of progress.**
- **Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.**
- **Continuous attention to technical excellence and good design enhances agility.**
- **Simplicity--the art of maximizing the amount of work not done--is essential.**
- **The best architectures, requirements, and designs emerge from self-organizing teams.**
- **At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.**

<https://agilemanifesto.org/principles.html>

We're losing the relay race

The... 'relay race' approach to product development...may conflict with the goals of maximum speed and flexibility. Instead a holistic or 'rugby' approach—where a team tries to go the distance as a unit, passing the ball back and forth—may better serve today's competitive requirements.

Hiroataka Takeuchi and Ikujiro Nonaka,
“The New New Product Development Game”,
Harvard Business Review, January 1986.



Scrum

Not an acronym, “A scrum is a **team pack** in Rugby, everybody in the pack acts together with everyone else to move the ball down the field”

A **small group** is responsible for **picking up the ball and moving it toward the goal**.



Schwaber, K. (October, 1995). Scrum development process. In *workshop proceedings of OOPSLA'95: Object-Oriented Programming, Systems, Languages & Applications, Austin, Texas*.

Scrum core concepts

Values: Commitment, Focus, Openness, Respect, and Courage

Separates two concerns

- In what sequence will work be done (PO)
- How much work will we commit to (Team)

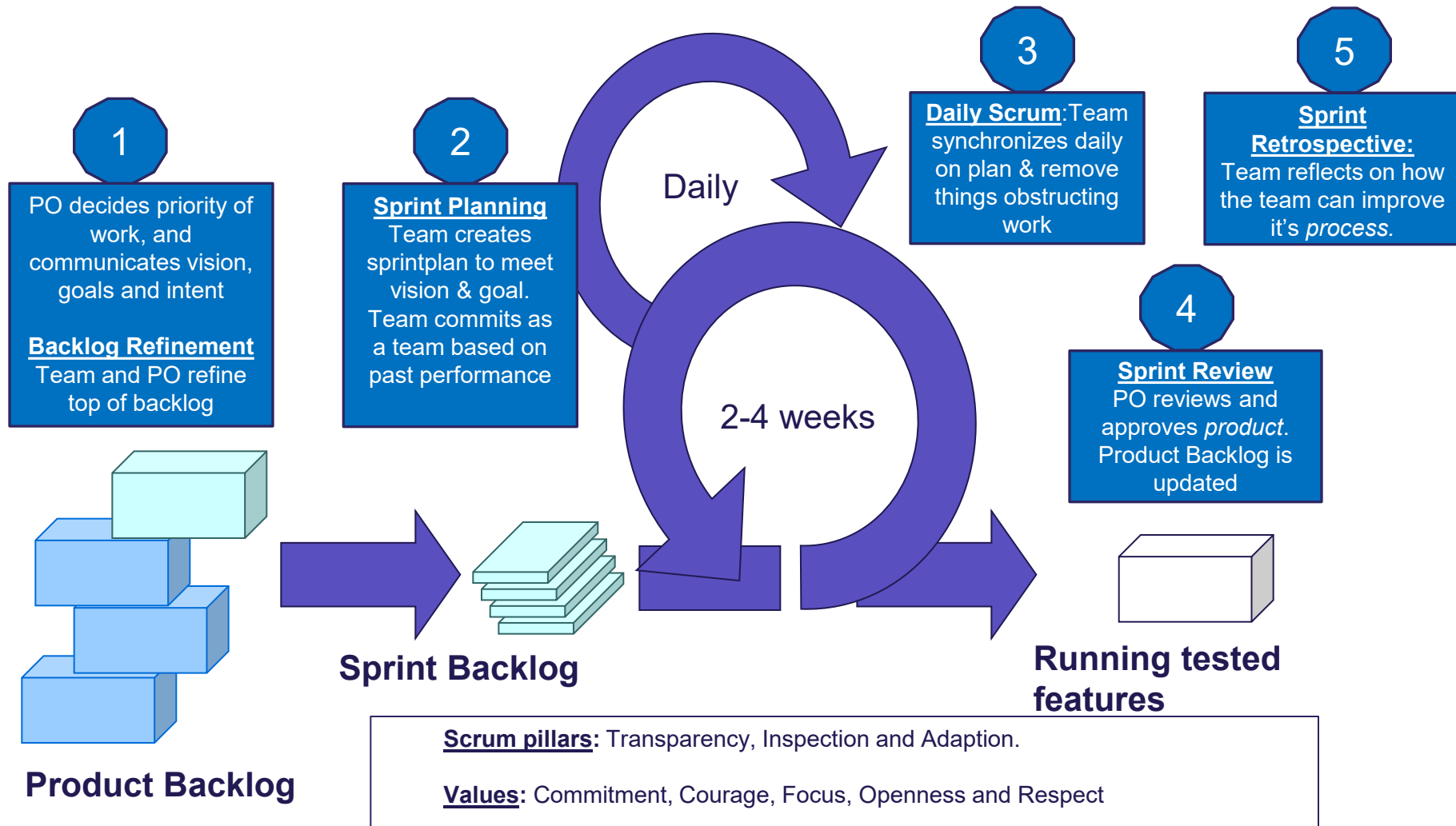
Transparency

- The emergent process and work must be visible to those performing the work as well as those receiving the work

Inspect and adapt

- More focus on goal/results than estimation precision or original plan

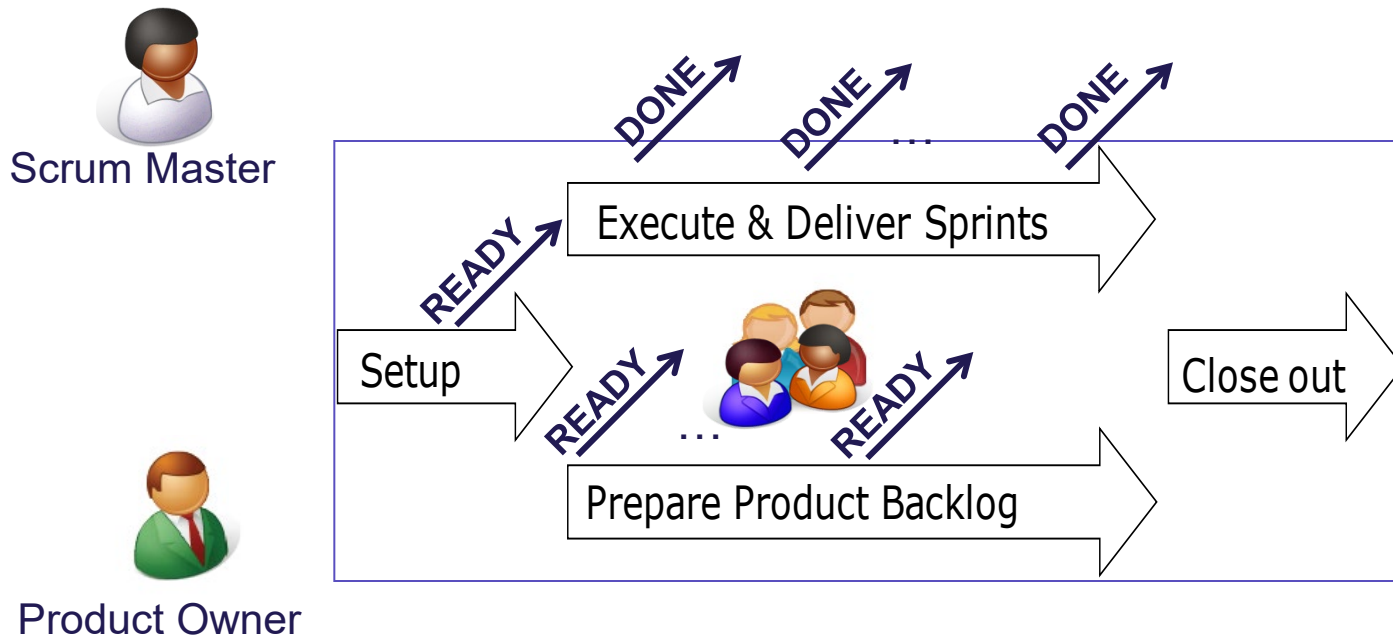
Scrum – Inspect & adapt based on transparency



Why may some people dislike Scrum?

Scrum has two concurrent tracks

Success depends on a balance in progress of the two tracks



When READY is well implemented a positive spiral emerge
When READY is ill implemented a negative spiral is started

Scrum Roles



Product Owner

- **Decision Maker: Order and content of Product Backlog**
 - Accepts or rejects work results
 - Responsible for *vision and profitability* of the product
 - Defines the features of the product, decides on release date and content - *release planning*.



Scrum Master

- **Facilitator of Scrum Process**
 - Ensures that the *Scrum process* is followed.
 - Enables close cooperation across all roles and functions
 - Removes barriers
 - Shields the team from external interferences
 - *NOT* a project manager



Team

- **Professionals who owns their own sprint plan**
 - Cross-functional, seven plus/minus two members
 - Selects the sprint goal and specifies work results
 - Has the right to do everything within the boundaries of the project guidelines to reach the iteration goal
 - Organizes itself and its work
 - Demos work results to the Product Owner

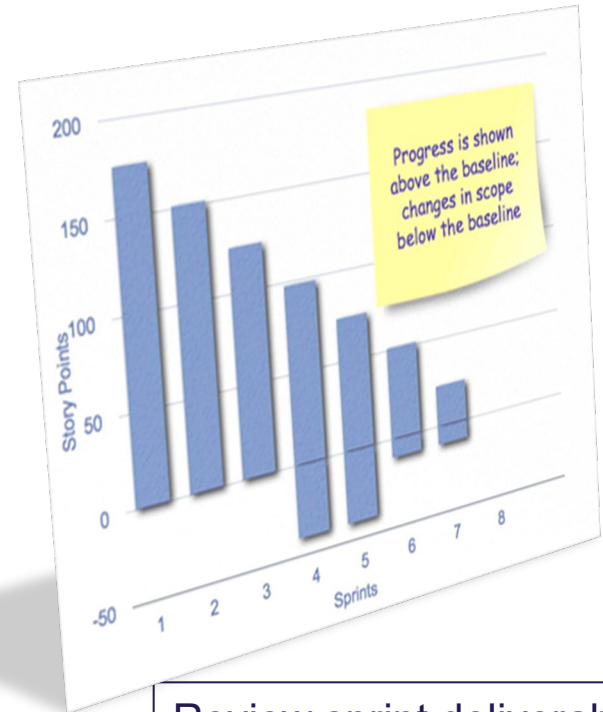
Scrum Product Owner activities



Customer

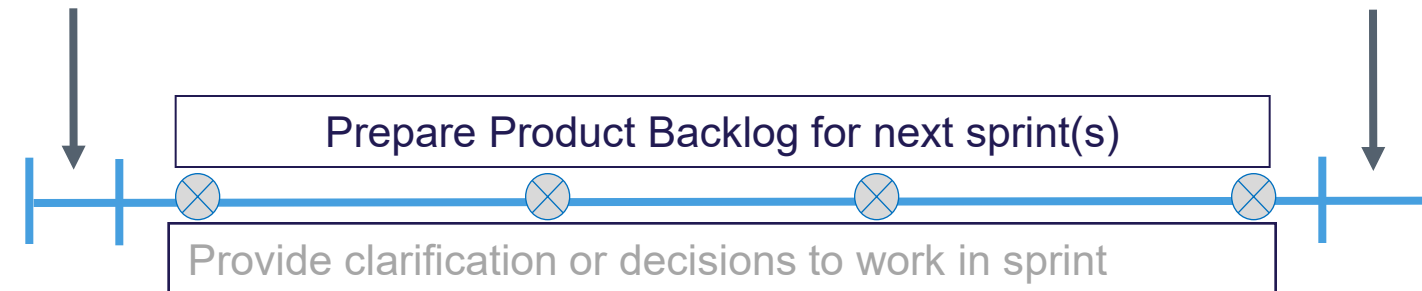


One person not a committee



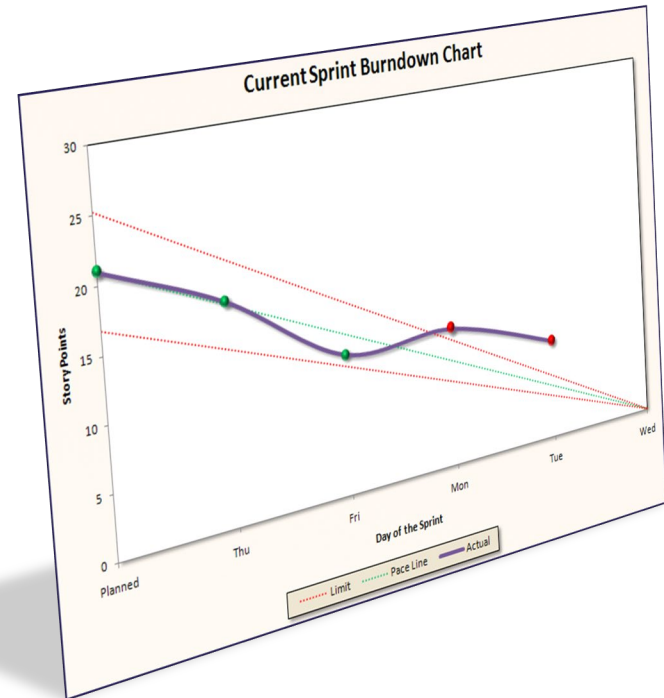
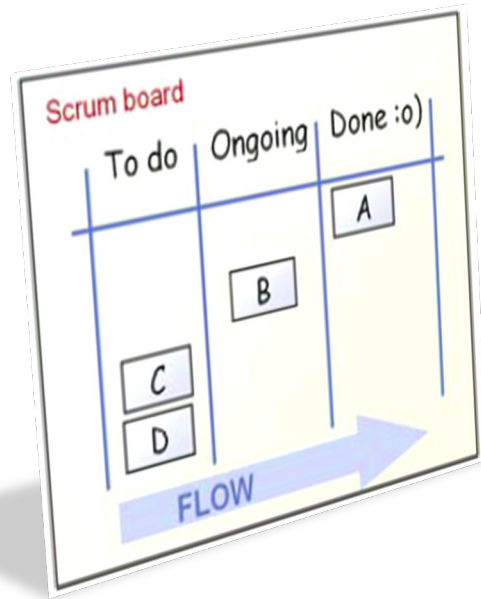
Prioritize Product Backlog

Review sprint deliverable



⊗ Refinement meetings

Scrum Master activities

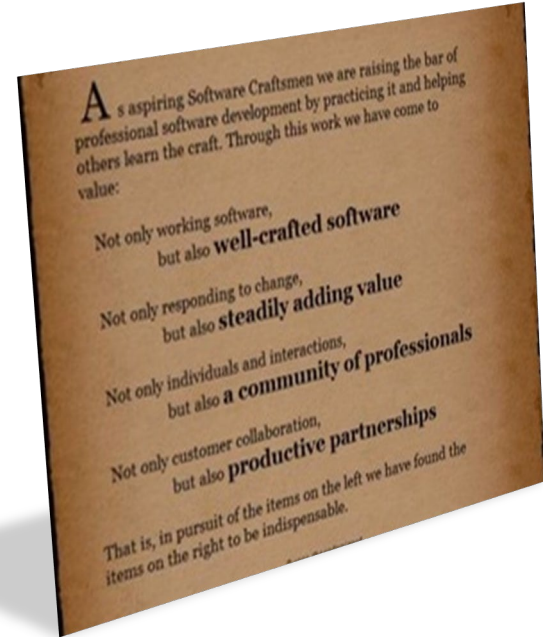


Facilitate Scrum meetings and coach development team and Product Owner



Remove impediments the team were unable to resolve
Protect team – help all to follow scrum process

Scrum Team activities



Sprint Planning

Sprint Review

Sprint Retro.

10% of team effort

Support Product Owner in clarification of work for future sprints

Deliver to sprint commitment

90% of team effort

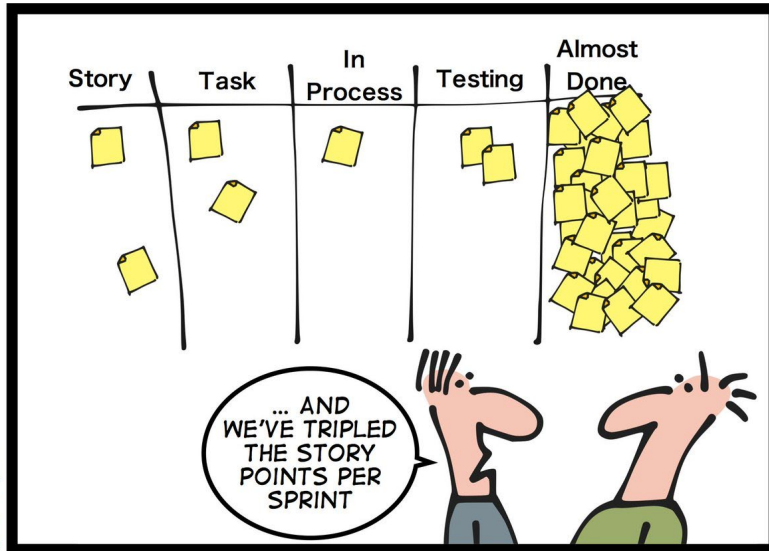
Scrum

Concept	Definition
Development team	A self-organizing group of software developers, which should be 7 +/-2 people. They are responsible for developing the software and other essential project documents.
Potentially shippable product increment	The software increment that is delivered from a sprint. The idea is that this should be 'potentially shippable' which means that it is in a finished state and no further work, such as testing, is needed to incorporate it into the final product. In practice, this is not always achievable.
Product backlog	This is a list of 'to do' items which the Scrum team must tackle. They may be feature definitions for the software, software requirements, user stories or descriptions of supplementary tasks that are needed, such as architecture definition or user documentation.
Product owner	An individual (or possibly a small group) whose job is to identify product features or requirements, prioritize these for development and continuously review the product backlog to ensure that the project continues to meet critical business needs. The Product Owner can be a customer but might also be a product manager in a software company or other stakeholder representative.

Scrum, contd.

Concepts	Definition
Daily scrum	A daily meeting of the Scrum team that reviews progress and prioritizes work to be done that day. Ideally, this should be a short face-to-face meeting that includes the whole team. Key questions: (1) What did you do Yesterday? (2) What will you do today? (3) Any impediments making it harder to reach the sprint goal?
Scrum Master	The Scrum Master is responsible for ensuring that the Scrum process is followed and guides the team in the effective use of Scrum. The Scrum Master should not be thought of as a project manager. Others, however, may not always find it easy to see the difference.
Sprint	A development iteration, usually 2-4 weeks long. Ends with a sprint review of work that meets the Done criteria (not done should not be presented).
Velocity	An estimate of how much product backlog effort that a team can cover in a single sprint. Understanding a team's velocity helps them estimate what can be covered in a sprint and provides a basis for measuring improving performance.

Challenges in Scrum



DOAD



Challenges in Scrum, contd.



Group exercises (10:15-12:00)

Exercise 1: Build your own Scrum (40%)

Exercise 2: Benefits and drawbacks of Scrum (20%)

- When is it a good idea to use Scrum?
- When is it better to use Waterfall?
- What are the benefits of Scrum?
- What are the drawbacks of Scrum or iterative and Incremental processes?

Exercise 3: Predictive versus adaptive planning and estimation of your semester project (40%)

- What does it mean that requirements in Scrum are emergent, and how is that similar or different to your semester project?
- How do we estimate in Scrum, and how do you estimate in your semester projects?
- Should your semester project be iterative (if yes, in how many iterations)?
- Should your semester project be incremental (if yes, in how many increments)?

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